

Lab 3

Component Modelling & Architectural Pattern Selection

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Problem Statement #58 — Digital Art Commission & Watermarking Portal

Architecture selection

We chose a Layered Architecture for the Digital Art Commission and Watermarking Portal.

The system sits in three layers. The Presentation Layer holds the Portal UI Component. The Business Layer holds the Order Manager, Watermarking Engine and Payment Service Components. The Data Layer holds the Asset Repository and Commission Database Components. Calls travel downward only, and the external Payment Gateway sits outside the system boundary behind the Payment Service.

Reason 1: the asset release rule needs a single enforcement point

FR-005 states that the watermark-free master becomes downloadable if and only if the gateway has authorized that milestone. Under a layered design every download request passes through the Order Manager, which reads the authorization record before it asks the Asset Repository for a URL. That leaves one code path to audit. Splitting these into independent microservices would give each service its own copy of payment state, and a stale replica would either release a paid asset early or keep a paying client locked out.

Reason 2: the workload does not justify distribution

The portal runs one commission workflow: submit a brief, upload drafts, approve, pay, download. Traffic is modest and the whole system is built and deployed by one person, so the operational cost of microservices buys nothing here. A plain client-server design, with the UI talking straight to the database, was rejected for the opposite reason. The watermarking obligation in FR-001 would then live in the client, where a user could bypass it and pull an unwatermarked draft.

Security advantage

The Presentation Layer has no route to the Data Layer. The Portal UI cannot address the Asset Repository and cannot name an object key; it can only call `ICommissionPortal`. The Asset Repository mints every download URL itself through `ISignedAssetUrl`, and only after the Order Manager has checked the authorization record, so a link that leaks stops working one hour after it was issued as NFR-001 requires. Card details never enter our layers at all. The Payment Service hands authorization to the external gateway over `IGatewayAuth` and keeps only the reference the gateway returns.

Performance benefit

Watermarking happens once per draft. When an artist uploads, the Watermarking Engine renders the diagonal overlay and writes the derivative to the Asset Repository through `IAssetStore`. Every later view of that draft serves the stored derivative, so the five-second budget in NFR-002 applies to the upload path alone rather than to each page load. The layers above the Data Layer hold no session state, so the Business Layer can be replicated behind a load balancer when the review cycle gets busy, and neither the UI nor the database schema has to change for that.